

[Secure Exit](#)SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT3-Debugging and Optimization

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QUESTIONS

Q1ATM Withdrawal – Exception
Handling Debugging
10 marks**Q2**Hospital Registration –
NullPointerException
Debugging
10 marks**Q3**E-Commerce Product Search –
Array Exception Debugging
10 marks**Q4**Bank Account – Race Condition
Debugging
10 marks**Q5**Railway Reservation –
Synchronization Debugging
10 marks

5/5 answered

Q1

ATM Withdrawal – Exception Handling Debugging

10
marks

An ATM application is used to process cash withdrawals. The following Java program contains programming errors.

Task:

1. Run the given program and identify the errors.
2. Debug and modify the program.
3. Submit the COMPLETE corrected Java program.
4. The corrected program must handle insufficient balance, zero withdrawal, invalid input, and unexpected errors.
5. Use appropriate try-catch-finally exception handling.
6. The corrected program must execute successfully for all valid test cases.

Faulty Program:

```
import java.util.Scanner;

class ATM {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter account balance: ");
int balance = sc.nextInt();

System.out.print("Enter withdrawal amount: ");
int amount = sc.nextInt();

try {
    int remaining = balance / amount;

    if (amount > balance) {
        System.out.println("Insufficient balance");
    } else {
        System.out.println("Withdrawal successful");
        System.out.println("Remaining balance = "
            + (balance - amount));
    }
}
catch (ArithmeticException e) {
    System.out.println("Transaction failed");
}

System.out.println("Transaction completed");
}
```

Expected correction:

The program must correctly validate the withdrawal amount before performing the transaction, handle division-by-zero and invalid input, and always display "Transaction completed" after exception handling.

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 class Atm
3 {
4     static class InsufficientBalanceException
5     {
6         InsufficientBalanceException(String
```

```
7      {
8          super(message);
9      }
10     }
11     public static void main(String[] args)
12     {
13         Scanner sc = new Scanner(System.in);
14         System.out.println("enter acc balanc
15         int balance = sc.nextInt();
16         System.out.println("enter the withdr
17         int amount = sc.nextInt();
18         try
19         {
20
21             if(amount>balance)
22             {
23                 throw new InsufficientBalance
24             }
25             if(amount<=0)
26             {
27                 throw new ArithmeticExceptio
28             }
29             int remaining = balance-amount;
30             System.out.println("withdrawl s
31             System.out.println("remainin
32
33         }
34         catch(ArithmeticException e)
35         {
36             System.out.println(e.getMessage(
37         }
38         catch(InsufficientBalanceException e
39         {
40             System.out.println(e.getMessage());
41         }
42         catch(Exception e)
43         {
44             System.out.println("invliad input
45         }
46         finally
47         {
48             System.out.println("transactuon
49         }
50     }
51 }
```

Test Input

```
2000  
1500
```

Output

```
enter acc balance  
enter the withdrawl amount  
withdrawl successful  
remaining balance =500  
transactuon completed
```

 Pending manual review

 Submitted

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